

Design note: the panel depth tier

Status: SIGNED (Mico, 2026-08-01, as part of the five-item agenda). Three commands: areg, xtivreg (fe), xtabond.

1. areg

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areg depvar indepvars [if] [in], absorb(catvar) [robust cluster()]
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Within-transform on absorb() groups (any variable, treated categorically by value), OLS on demeaned data, coefficients identical to including group dummies; dof: $N - K - G$ (G = number of absorbed groups), so classical/robust SEs use the dummy-equivalent df. SEs through the vce module (its fourth act). VERIFICATION: internal exactness against tea's own regress with the dummy set — coefficients equal to machine precision, SEs equal after the df correction.

2. xtivreg, fe

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xtivreg depvar [exog] (endo = instruments), fe [robust cluster()]
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Within transform (panel demeaning) of every variable, then 2SLS on the demeaned system; df accounts for the N_g absorbed means (Stata's small-sample convention: $df = N - K - N_g$). fe only; re is staged. VERIFICATION: internal exactness against manual demeaning + ivregress 2sls in tea itself, plus a numpy referee.

3. xtabond

```
xtabond depvar [indepvars], lags(1) [maxldep(#) robust twostep]
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Arellano-Bond difference GMM, release scope: lags(1) (one lag of the dependent variable), strictly exogenous indepvars instrumented by themselves in differences, GMM-style (non-collapsed) instruments for the lagged dependent variable: at time t , levels $y_{i,1} \dots y_{i,t-2}$, optionally capped by maxldep(). One-step estimator with the Arellano-Bond H matrix (tridiagonal 2,-1) as the initial weight; twostep option uses the one-step residual-based optimal weight. vce: conventional one-step SEs, or robust = panel-clustered sandwich (Stata's vce(robust)); Windmeijer correction for two-step is STAGED (two-step robust SEs print a note). Sargan statistic reported from the two-step (or one-step homoskedastic) J. AR(1)/AR(2) tests staged. VERIFICATION: an independent numpy GMM referee implementing the same formulas on the same simulated panel (no reference implementation is installed; the referee is hand-rolled and the formulas are cross-checked against Arellano-Bond 1991), plus rho recovery on a known DGP.

Shared

Panel machinery uses xtset (ts_panel/ts_time). All three post Estimates with the standard machinery, honor if/in via the house Sel idiom where applicable, and are exercised by tolerance-first tests where estimators are iterative or sampling-based, exact goldens where deterministic (areg/xtivreg are deterministic: exact goldens).